Truly Mobile Sub-Terahertz Communications Beyond 6G via Just-in-Time IMU-Assisted Beam Management

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ABSTRACT

Sub-Terahertz (sub-THz) communication is a candidate technology for beyond-sixth-generation (beyond-6G) wireless networks, promising up to Terabit-per-second data rates and microsecond-level latencies. However, the highly directional sub-THz beams required to overcome severe path loss make maintaining reliable beam alignment under mobility extremely challenging. Existing *reactive* in-band and out-of-band sensor-assisted beam management strategies are often insufficient to prevent outages caused by unpredictable user motion. Progress is further challenged by the absence of comprehensive evaluation platforms that jointly account for: (a) realistic user mobility, (b) real-time sensor data, and (c) accurate characteristics of sub-THz hardware. To address these challenges, we introduce: (i) *Just-in-Time (JIT)*, a *hybrid proactive and reactive beam management algorithm* that fuses RF and inertial measurement unit (IMU) data to preemptively realign the link before failure, and (ii) *MobileTHz*, an integrated simulation and Hardware-in-the-Loop platform for mobile sub-THz systems. Benchmarked against an in-band hierarchical search and a sensor-assisted reactive algorithm, JIT significantly improves mean capacity, reduces variance, and maximizes link availability while incurring a signaling overhead of less than 0.4% of the channel capacity. Together, JIT and MobileTHz advance efficient, resilient, and truly mobile sub-THz communication systems beyond 6G.

Introduction

The trajectory of wireless communications development is heavily defined by a growing demand for higher reliability and capacity, lower latency, and ubiquitous connectivity[1–3]. Despite an initially relatively slow rollover of fifth-generation (5G) wireless network deployments globally, in recent years, the use of high-definition video streaming, cloud-based services (including artificial intelligence (AI) systems), and the Internet of Things (IoT) has continued to accelerate[2,4]. According to multiple predictions and analyses, future 6G-and-beyond applications such as immersive extended reality (XR), large-scale digital twinning, connected robotics, and high-rate non-terrestrial networks will place demanding requirements on wireless infrastructure. These applications may ultimately require peak data rates reaching the Terabit-per-second (Tbps) scale, coupled with latencies on the order of microseconds[2,5–8].

Multiple ongoing R&D efforts worldwide aim to improve the efficiency of existing wireless systems through further network densification[9], the deployment of massive multiple-input and multiple-output (MIMO) systems[10], the utilization of Orbital Angular Momentum (OAM)[11], novel modulation and coding schemes[12], integrated sensing and communication (ISAC)[13], and more efficient AI-native air interfaces[14], among others. However, the key performance indicators for wireless systems, such as capacity and latency for communication and resolution for sensing, are tightly coupled to the frequency and bandwidth at which the wireless system operates. Hence, the availability of sufficient spectrum resources becomes essential. Meanwhile, the spectrum allocated for existing 5G networks spans the sub-6 GHz band (Frequency Range 1, FR1), mid-bands between 7 GHz and 15 GHz (sometimes referred to as FR3[15]), and millimeter-wave (mmWave) bands below 100 GHz (such as FR2 and FR2-2). These resources, which the forthcoming first 6G releases are expected to share[16], are projected to become a significant bottleneck[17–19].

To address this challenge, the research community is actively exploring the sub-Terahertz (sub-THz) spectrum, spanning from 100 GHz (0.1 THz) to 1 THz (also referred to as "higher mmWave" bands[20]), as a possible next frontier for wireless communication systems beyond 6G[1,2,8,18,19,21]. The sub-THz spectrum particularly offers large contiguous blocks of unallocated bandwidth, up to tens or even hundreds of gigahertz of usable spectrum[3,22]. Efficiently utilizing such a wide spectrum will provide one of the essential prerequisites to achieve Tbps-level data rates in future wireless systems[7].

Although the sub-THz spectrum offers a solution to bandwidth scarcity, the properties of the sub-THz channel present many challenges. At these frequencies, free-space path loss presents a significant challenge[5,21,23]. Additionally, sub-THz signals exhibit poor diffraction and penetration capabilities, making them extremely sensitive to blockage by common obstacles, such as a human hand or body, which are sufficient to interrupt a communication link[19,24]. *To counteract these issues, sub-THz systems must employ highly directional, high-gain beams on both the transmitter (Tx) and the receiver (Rx) sides[21].*

Notably, with the evolution of sub-THz technology in recent decades[3,25], the viability of sub-THz communications is rapidly moving from theoretical speculation to prototyping, demonstration, and practical system design. Recent experimental milestones include offline systems achieving 1.0488 Tbps in the 330–500 GHz band and long-distance links such as 18 Gbps over 30.2 km[23]. Real-time software-defined radios have also demonstrated 33 Gbps data rates in the 135 GHz sub-THz band[26,27], among many other recent advancements in sub-THz hardware.

However, while for **stationary** links (e.g. backhaul or fronthaul setups) high-rate data exchange using highly-directional sub-THz beams has already been experimentally demonstrated via various testbeds[28–30], building reliable **mobile** sub-THz communication systems remains an open challenge with only a few successful hardware trials reported to date[31,32]. The key difficulty here is to ensure continuous alignment of narrow Tx and Rx sub-THz beams in the presence of unpredictable node mobility (both translation and rotation)[32–34]. A deviation of just a few degrees between the Tx and Rx beams can lead to complete loss of the received signal, immediately breaking the connection and making it difficult to coordinate Tx and Rx actions to promptly realign their beams. *Hence, developing a reliable mechanism to ensure close-to-perfect alignment of sub-THz beams is one of the key challenges in converting mobile sub-THz communications from concept to realization[6,32,35].*

Maintaining beam alignment under mobility, commonly referred to as part of beam management, is a key bottleneck for mobile sub-THz networks[34,36,37]. Conventional **purely in-band** approaches used in modern networks, such as Kalman-filter-based tracking of Rx power, signal-to-noise ratio (SNR), bit error rate (BER), and other RF measurements[38–41], demonstrate reliable performance for smooth, linear node motion but struggle with abrupt changes, such as unpredictable rapid device rotation by a human user[34]. This limitation has recently motivated a shift toward developing new **out-of-band** strategies that also incorporate auxiliary sensor data into the beam management[6,7,16,19,24]. Particularly, infrastructure-side sensors (radar, LiDAR, cameras) can track user position[37,42], while built-in on-device Inertial Measurement Units (IMUs) provide low-latency motion estimates directly at the user equipment (UE). Fusing IMU and RF data has been shown to outperform either source alone[35,43–46]. *Still, the principal challenge remains, once the Tx and Rx sub-THz beams get misaligned, realigning them quickly in a mobile environment is difficult without an active Tx-Rx link, even with the use of auxiliary data (e.g., from IMUs).*

Notably, the progress in this field is further limited by a gap in comprehensive end-to-end sub-THz-specific evaluation platforms. On one side, existing sub-THz simulators, including TeraSim[47], NYUSIM[48], SiMoNe[49,50], and TeraMIMO[51,52], primarily focus on carefully modeling static or quasi-static point-to-point links and are not inherently designed for real-time beam tracking under mobility[32]. Even when modeling mobility, it is usually accounted for with a low-complexity first-order model: limited straight trajectories, no realistic acceleration, complex non-linear device rotations, etc. On the other hand, although various hardware alignment platforms exist for lower frequency bands[53–55], most of them do not account for the unique constraints of sub-THz hardware and data links.

To address the gap mentioned above, in this article, we design, develop, and comprehensively evaluate (both through simulations and a hardware testbed) a *novel Just-in-Time (JIT) mobile sub-THz communication system implementing a new IMU-assisted beam management protocol*. The presented solution operates in both a reactive regime, facilitating fast beam realignment once the link is broken, and a **proactive** regime, predicting possible link failure and coordinating Tx and Rx efforts to prevent it or, when prevention is not possible, minimize the duration of misalignment. We further present *MobileTHz*, a new hardware and software platform, implementing this proposed beam management procedure for mobile sub-THz communications. Finally, we utilize the MobileTHz platform to compare the performance of our proposed solution against state-of-the-art IMU-assisted and non-IMU-assisted beam management approaches.

Our analysis demonstrates that the proposed JIT solution consistently outperforms baseline sub-THz beam management techniques in terms of achievable capacity, lower capacity variance, and greater link availability (*by up to 1–2 orders of magnitude*, depending on the scenario). We therefore conclude that implementing such a new hybrid reactive+proactive IMU-assisted beam management approach can become one of the essential building blocks for truly mobile, reliable, and efficient sub-THz communications toward beyond-6G wireless networks.

"Just-in-Time" Proactive Sub-THz Beam Management

Conventional millimeter-wave beam management strategies typically operate reactively, initiating recovery procedures only after signal degradation or link failure is detected[34,36]. This approach is challenging in sub-THz networks due to the reliance on high-gain antennas[21]. While essential for closing the link budget, narrow beams create mobility constraints: they quadratically expand the spatial search space while shrinking the beam residence time (the window where a moving node stays aligned).

The proposed JIT hybrid proactive+reactive sub-THz beam realignment approach, illustrated in Fig. 1, bridges this gap by leveraging IMU sensor data on a mobile UE to provide the access point (AP) with real-time motion awareness. The algorithm fuses RF and IMU data to continuously monitor the UE's deviation from the beam center. A trigger boundary, set inside the antenna's Edge of Coverage (EoC), determines when preemptive action is required. When the projected deviation

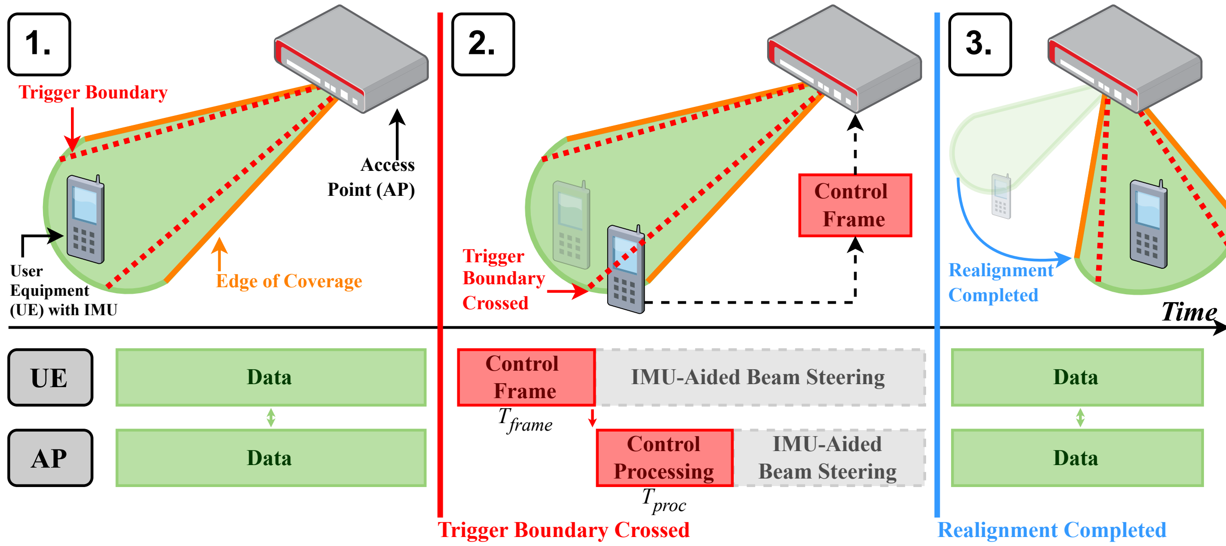


Figure 1. Overview of JIT proactive beam management. (1) Steady-state operation with continuous RF and IMU monitoring. (2) Upon crossing the trigger boundary, the UE transmits a control frame (T_{frame}) containing its kinematic state *just before* exiting the Edge of Coverage; the AP processes it (T_{proc}) and both nodes initiate beam steering. (3) Realignment completes and the data link is restored. The bottom timeline shows the UE and AP activity during the procedure.

crosses this boundary, the UE transmits a compact control frame containing its kinematic state to the AP before the link is lost. Then, both nodes initiate coordinated IMU-aided beam steering, completing realignment, or, when prevention is not possible, facilitating a deterministic prompt recovery. Additionally, JIT leverages the sub-THz spectrum to transmit control frames with microsecond-level latency, allowing the system to react to mobility events within the short beam residence time.

To validate this proposal, we implemented the JIT solution using the MobileTHz platform, encompassing both a high-fidelity software simulator and a hardware platform (detailed in the *Methods* section). We then performed a comparative study, benchmarking the performance of the JIT beam realignment against two conventional reactive approaches: the Hierarchical Search algorithm and the IMU-Assist algorithm. The details of the study, including experimental assumptions and system parameters, are summarized in *Methods*, while the key results of our evaluation are discussed in the *Results* section.

Results

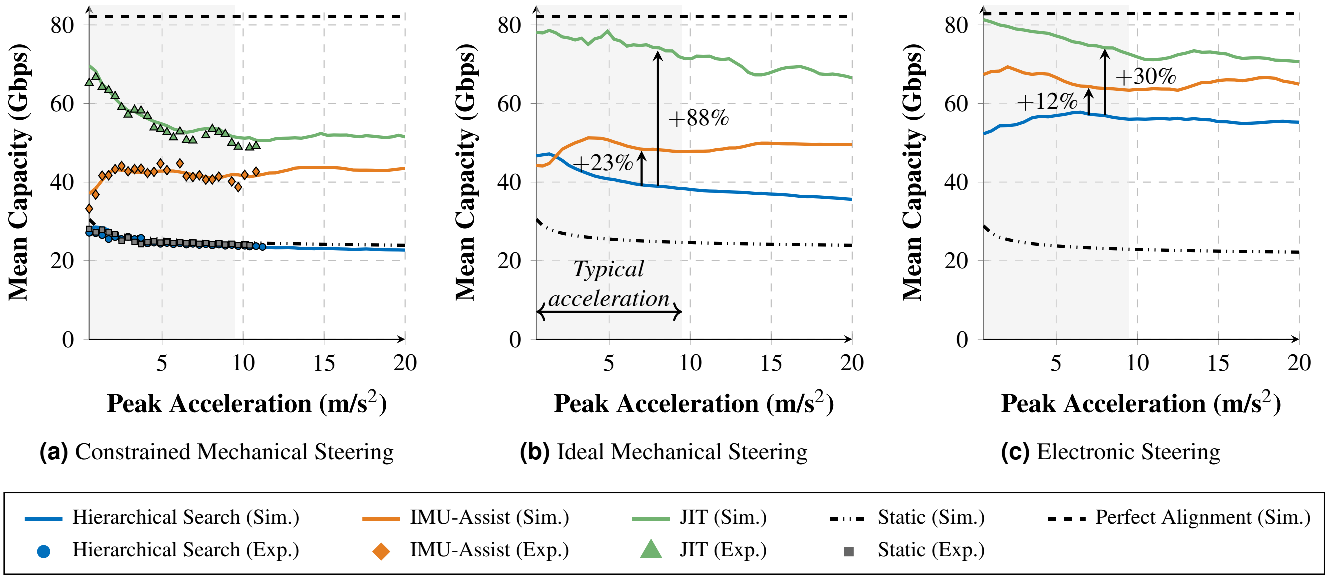
We benchmark JIT against four reference schemes. The first two are: (i) a state-of-the-art beam recovery approach (without IMU assistance); and (ii) a reactive IMU-assisted beam alignment (leveraging IMU data for beam realignment, but without JIT signaling). For illustration, we also model (iii) a pure static (no realignment) baseline approach used in stationary sub-THz communication systems; and (iv) a perfect alignment setup (a non-constructive Oracle-based optimal setup with continuous perfect beam alignment and no signaling overheads regardless of the nodes' mobility). The latter also serves as a theoretical upper bound for any constructive beam realignment approach.

The key performance indicators when comparing these solutions include Mean Capacity, Capacity Coefficient of Variation (CoV), Link Availability, and Signaling Overhead (further discussed in Methods). Each group of results below illustrates the impact of different key input parameters: specifically, the separation distance between the nodes, the UE translational acceleration, UE rotational acceleration, directivity of the employed sub-THz antenna system, and various hardware constraints.

Notably, for the latter, the performance levels for different beam alignment schemes (both the proposed JIT and the reference ones) are specifically evaluated across three representative beam steering models. These models reflect different types of steering mechanisms that can be possibly used in practical sub-THz communications (see Table 1). Specifically, *Constrained Mechanical* mirrors our laboratory testbed and anchors Hardware-in-the-Loop (HIL) validation; *Ideal Mechanical* represents the upper bound of mechanical actuation for use cases such as backhaul and UAV links with high-speed rotation platforms; and *Electronic* represents phased arrays envisioned for future mobile sub-THz access networks, extending state-of-the-art mmWave UE and AP phased antenna arrays to the sub-THz band.

Effects of Translational Acceleration on Beam Alignment

We consider an AP and UE initially separated by $D_{initial} = 3$ m, an indoor short-range link[56]. After a brief stationary period, the UE moves $D_{traversed} = 1.5$ m in a random direction relative to the AP, modeling human-scale mobility, such as walking between workstations. A representative received power profile for this scenario is shown in Fig. 3a.



(a) Constrained Mechanical Steering **(b)** Ideal Mechanical Steering **(c)** Electronic Steering

Figure 2. Translational mobility evaluation ($D_{initial} = 3$ m, $D_{traversed} = 1.5$ m) across three steering models. HIL markers in (a) validate simulation with MAPE of 1.80–3.13% and NRMSE of 2.99–3.82%. Percentage annotations in (b, c) are relative to Hierarchical Search at the marked acceleration.

Under the Constrained Mechanical model (Fig. 2a), slow mechanical steering imposes the harshest penalty on every realignment attempt. Under these constraints, Hierarchical Search frequently fails to complete its scan before the UE moves away, causing it to achieve 5–7% worse mean capacity than a static, non-steered link across the acceleration range. The IMU-Assist algorithm mitigates this by narrowing the angular search space using on-device motion estimates, improving mean capacity over Hierarchical Search by up to 92% at 20 m/s^2. JIT further extends this advantage by preemptively sharing kinematics with the AP, achieving up to an 88% improvement over IMU-Assist at low accelerations. At the highest mobility, IMU-Assist and JIT reach 50% and 69% of the theoretical maximum, respectively, confirming that sensor-informed algorithms can make mechanically constrained hardware viable for mobile links where a purely reactive approach cannot.

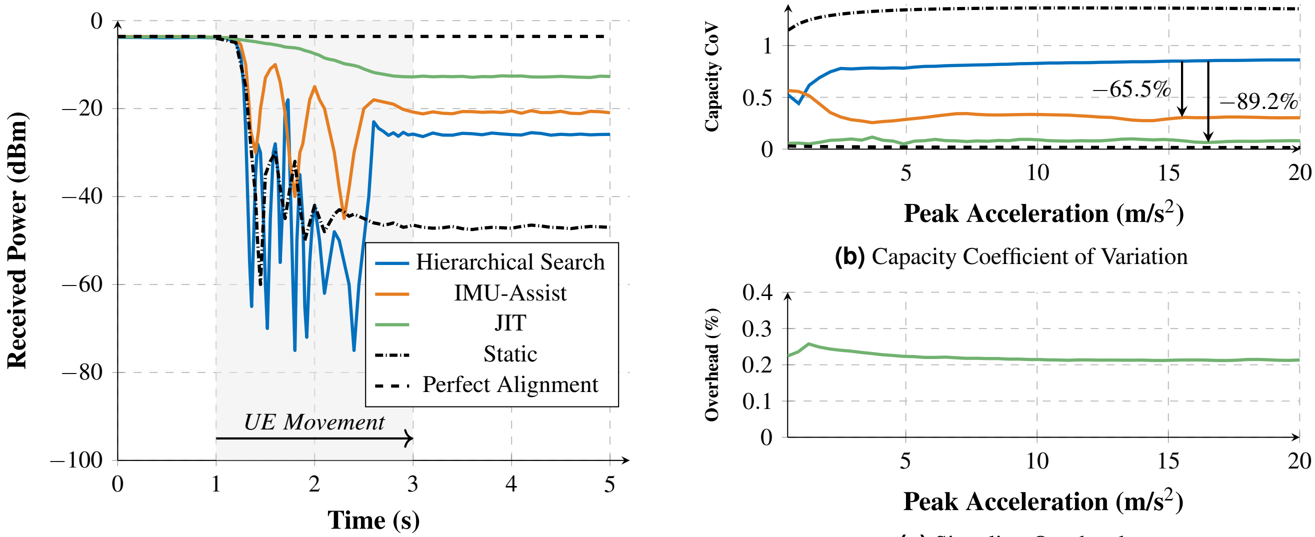
To validate these simulation results, we conducted Hardware-in-the-Loop experiments across all four algorithm configurations. The experimental data closely matched the simulation, with Mean Absolute Percentage Error (MAPE) values of 1.80–3.13% and Normalized Root Mean Square Error (NRMSE) of 2.99–3.82%, confirming that the MobileTHz engine captures the physical constraints of the steering hardware. The subsequent analysis therefore uses simulation to explore steering models beyond our laboratory equipment.

Under the Ideal Mechanical model (Fig. 2b), faster rotary movements impose a lower opportunity cost than the Constrained Mechanical model. Hierarchical Search now outperforms the static reference by at least 75%, since the faster hardware can complete enough scan iterations within the beam residence time to recover alignment. IMU-Assist provides further gains of up to 39% at 20 m/s^2, though it slightly underperforms at 0.5 m/s^2 due to overshooting from aggressive motion prediction (discussed in Methods). JIT delivers the largest improvement across the full range, narrowing the gap to the theoretical maximum by synchronizing both nodes rather than relying on the AP's reactive scan alone. This result demonstrates that even when hardware is fast enough for reactive recovery to succeed, proactive coordination still yields substantial gains by eliminating unnecessary search overhead.

With the Electronic model (Fig. 2c), where steering latency is negligible, the performance spread narrows: Hierarchical Search plateaus at 55.3 Gbps, IMU-Assist at 63.4 Gbps, and JIT at 72.6 Gbps. Yet JIT still retains a 12% advantage over IMU-Assist and 30% over Hierarchical Search, showing that the search-space penalty of reactive methods persists even when hardware is no longer the bottleneck.

Although mean capacity summarizes maximum link performance, it masks temporal variance, making it insufficient as a sole metric for assessing mobile link robustness. A single mobility event is illustrated in Fig. 3a, showcasing the power variations over time. Hierarchical Search experiences the most power oscillations during movement, while the IMU-Assist algorithm decreases temporal variance as well as the magnitude of link loss. JIT outperforms the other algorithms, providing a stable link and the highest final received power of −12.7 dBm.

The CoV of the capacity across the three algorithms is quantified in Fig. 3b as peak acceleration increases under the Ideal



(a) Captured Rx Power across entire movement scenario

(b) Capacity Coefficient of Variation

(c) Signaling Overhead

Figure 3. Ideal Mechanical, translational scenario ($D_{initial} = 3$ m, $D_{traversed} = 1.5$ m, 3.0 m/s^2 peak). (a) Single-run trace at 3.0 m/s^2; shaded region indicates UE displacement interval. (b, c) Aggregated over the full acceleration sweep.

Mechanical model. The IMU-Assist algorithm reduces CoV by up to 66% compared to Hierarchical Search, demonstrating that IMU integration improves not only mean capacity (as shown in Fig. 2b, where IMU-Assist achieved up to a 39% increase) but also reduces capacity variance. However, single-node motion compensation alone is insufficient to achieve a consistently low-variance link. JIT shows that capacity variance can be further reduced when both nodes share mobility awareness, reducing CoV by up to 89.2% compared to Hierarchical Search and by 23.7% compared to IMU-Assist.

Overhead, defined as the potential channel capacity consumed by JIT transmission, is quantified in Fig. 3c. Overhead increases up to 1.3 m/s^2, the point at which the UE consistently moves beyond the AP's EoC causing more realignment events. At higher accelerations, however, overhead decreases as the AP's look-ahead strategy results in fewer intermediate realignments before the UE reaches its final position. Overall, JIT improves mean capacity and reduces link variance while consuming at most 0.26% of the total channel capacity, demonstrating its efficiency in high-mobility regimes.

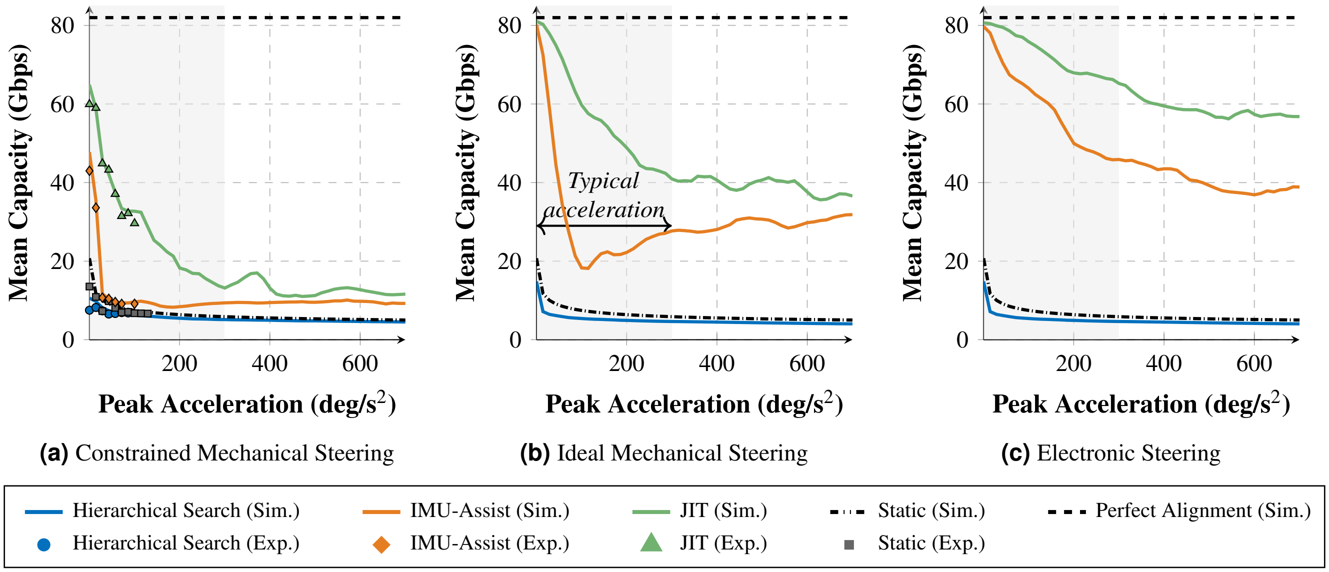
Effects of Rotational Acceleration on Beam Alignment

Rotational motion often presents the greatest challenge to beam management, as typical angular rates from rotation can be significantly higher than those induced by translation[43] (see also Supplementary Figs. 1 and 2). Unlike translation, where the geometric Line-of-Sight (LoS) angle changes at both nodes and both must steer to compensate, pure rotation alters only the UE's orientation, and the AP's LoS remains unchanged. The correct response is therefore for the UE to counter-rotate while the AP holds its position. Reactive algorithms, relying solely on RF power drops, cannot distinguish rotation from translation and default to searching at both nodes. Under pure rotation, this causes the AP to actively steer *away* from the correct direction, turning the recovery mechanism itself into a source of misalignment.

This behavior is quantified in Fig. 4b, where Hierarchical Search's capacity degrades to 5–6 Gbps, consistently falling below the static reference across the full acceleration range. The IMU-Assist algorithm addresses this failure at one node: the UE uses on-device sensor data to identify the motion as rotation and counter-rotates to maintain alignment rather than initiating a scan, recovering capacity to approximately 30 Gbps. However, this correction is local to the UE. The AP, unaware that the UE has already compensated, may still detect a transient power drop and trigger its own reactive scan, partially undoing the UE's correction. This coordination gap caps IMU-Assist at around 36.5% of the theoretical maximum. JIT closes this gap, sustaining capacity above 55 Gbps up to 120 deg/s^2. Even at 300 deg/s^2, representative of the upper range of human rotational acceleration (see Supplementary Figs. 1 and 2), JIT maintains a 47% advantage over IMU-Assist.

The Electronic model (Fig. 4c) also demonstrates that Hierarchical Search remains below the static reference, since fast steering cannot correct a misdirected search. IMU-Assist benefits substantially, settling around 38 Gbps at 700 deg/s^2, while JIT sustains above 57 Gbps across the full range by ensuring the AP holds position rather than scanning.

Under experimental constraints (Fig. 4a), all algorithms degrade more rapidly than in the idealized scenarios because the rotary stages cannot complete the required angular correction within the beam residence time. Hierarchical Search remains at 7–8 Gbps throughout, consistently below the static reference. IMU-Assist falls from 43 Gbps to under 10 Gbps by 100 deg/s^2, while JIT drops from 60 Gbps to 30 Gbps over the same interval, retaining a 3× capacity advantage, demonstrating that hybrid

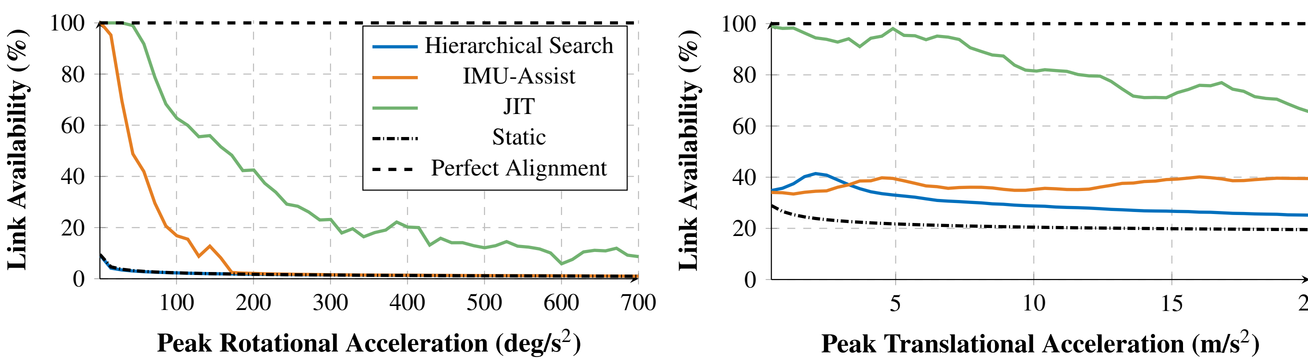


(a) Constrained Mechanical Steering **(b)** Ideal Mechanical Steering **(c)** Electronic Steering

Figure 4. Rotational mobility evaluation ($D_{initial} = 3$ m, 45° displacement) across three steering models. Hierarchical Search falls below the static reference in all cases due to misdirected AP scanning. HIL validation in (a): MAPE of 3.88–4.68% for JIT and IMU-Assist; higher for Hierarchical Search (11.66%) due to low absolute capacity.

proactive+reactive coordination can extend the usable rotational range of mechanically steered systems well beyond what reactive algorithms support. Experimental results track simulation with MAPE of 3.88–11.66% and NRMSE of 5.19–17.5%, with the higher errors concentrated in Hierarchical Search, where low absolute capacities amplify small deviations. JIT and IMU-Assist show closer agreement (MAPE of 3.88% and 4.68%), validating the MobileTHz engine under rotational dynamics.

While capacity CoV (Fig. 3b) quantifies variance over the entire scenario, it does not indicate what fraction of time the link actually sustains the data rates that sub-THz systems are designed to deliver[1,7]. Link availability addresses this by measuring the fraction of time SNR exceeds 18.8 dB (derived in *Evaluation Metrics* subsection). A system may exhibit low CoV yet poor link availability if it sustains a steady but weak signal below the modulation threshold.



(a) Link Availability vs. Peak rotational acceleration **(b)** Link Availability vs. Peak translational acceleration

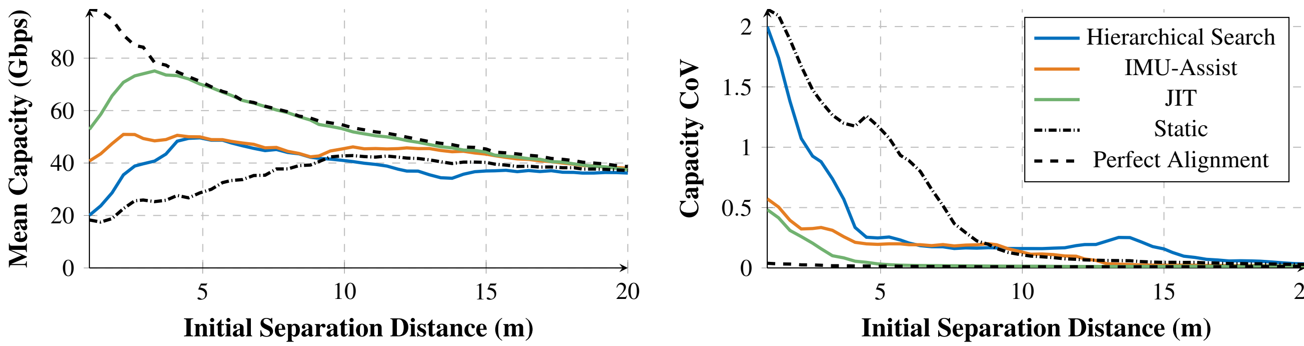
Figure 5. Link availability (SNR $\geq$ 18.8 dB, 64-QAM threshold) under Ideal Mechanical vs. peak rotational (a) and translational (b) acceleration. Complements Figs. 2b and 4b by showing usability rather than mean capacity.

The link availability under the Ideal Mechanical model is quantified in Fig. 5. In the rotational case (Fig. 5a), Hierarchical Search falls below 5% above 15 $\mathrm{deg/s^2}$, offering no improvement over the static reference. IMU-Assist begins at 100% but drops below 50% by 43 $\mathrm{deg/s^2}$ and converges to the Hierarchical Search floor above 170 $\mathrm{deg/s^2}$, as the AP's uncoordinated scanning negates the UE's local correction. JIT holds 100% up to 29 $\mathrm{deg/s^2}$, remains above 80% at 72 $\mathrm{deg/s^2}$, and maintains a usable link (>40%) up to 200 $\mathrm{deg/s^2}$. These results demonstrate that compared to single-node correction, end-to-end node synchronization significantly improves link availability under human-scale rotational accelerations.

In the translational case (Fig. 5b), Hierarchical Search and IMU-Assist plateau between 25–40% across the full acceleration range, neither reliably supporting 64-QAM. JIT's availability declines gradually at higher accelerations as the beam residence time shrinks, but remains above 65% at 20 m/s^2. The reactive methods, by contrast, stay flat at 25–40% regardless of acceleration, as their search cannot converge before the UE exits the beam.

Effects of Link Geometry on Mobile Performance

We next isolate the effect of link geometry by fixing the motion profile and varying two parameters: initial separation distance (1 m to 20 m) during a translational pass-by (1.5 m displacement, 3.0 m/s^2 peak acceleration) and total angular displacement (1° to 120°) during a rotational turn.



(a) Mean Capacity vs. Separation Distance

(b) Capacity CoV vs. Separation Distance

Figure 6. Effect of initial AP–UE separation (1–20 m) on a fixed translational pass-by ($D_{traversed} = 1.5$ m, 3.0 m/s^2 peak), Ideal Mechanical. Performance converges beyond ∼10 m as path loss, not alignment, becomes the limiting factor.

Mean capacity as a function of initial separation distance (1 m to 20 m) is shown in Fig. 6a. At close range, the fixed 1.5 m displacement corresponds to a large angular change relative to the AP, generating angular velocities that overwhelm reactive tracking. This is evident at 1 m, where Hierarchical Search achieves only 20 Gbps, slightly above the static reference of 18 Gbps, while IMU-Assist and JIT reach 41 Gbps and 53 Gbps, respectively. A crossover occurs at ≈4.5 m: beyond this distance, angular rates decrease enough for Hierarchical Search to recover the link, reaching 49 Gbps. The performance gap narrows further beyond 10 m as path loss reduces the link budget, and all algorithms converge to 37 Gbps to 39 Gbps by 20 m, where alignment accuracy is no longer the limiting factor.

The capacity CoV (Fig. 6b) complements this view. At 1 m, Hierarchical Search exhibits a CoV of 2.0, indicating highly erratic capacity, while IMU-Assist and JIT reduce this to 0.57 and 0.48. As separation increases to 4.5 m, JIT achieves near-zero CoV (0.047) while Hierarchical Search and IMU-Assist settle at 0.25 and 0.20, respectively. JIT maintains CoV below 0.02 across all distances beyond 5 m, confirming stable link behavior regardless of link geometry.

For the rotational sweep (Fig. 7a), Hierarchical Search drops from 81 Gbps to under 9 Gbps by 35° and settles at ≈5 Gbps, again falling below the static reference due to misdirected AP scanning. IMU-Assist maintains 81 Gbps through 10° but degrades to 34 Gbps by 40° as the AP's uncoordinated scanning erodes the UE's correction. JIT sustains above 55 Gbps through 50° and remains above 40 Gbps at 120°, maintaining a capacity advantage of at least 2× over IMU-Assist across the full angular range.

Link availability over the same sweep is quantified in Fig. 7b. Hierarchical Search falls below 50% by 11° and under 5% by 20°. IMU-Assist holds 100% through 10° but drops below 50% by 28° and converges toward Hierarchical Search's floor above 90°. JIT maintains 100% through 30° and remains above 40% at 120°, extending the usable angular range by approximately 3× compared to Hierarchical Search. This gain requires additional signaling, quantified in Fig. 7c. JIT overhead rises from 0.01% at 1° to 0.23% at 35° as the algorithm triggers more frequent realignment updates, then plateaus at ≈0.39% for angles above 90°. Even at maximum rotation, this represents a negligible fraction of total channel capacity.

Impact of Antenna Directivity on Combined Motion

The preceding sections evaluated each motion type independently. We now combine translational (1.5 m displacement) and rotational (90° turn) motion simultaneously, with acceleration escalating from (0.5 m/s^2, 0.5 deg/s^2) to (5 m/s^2, 200 deg/s^2). We evaluate four antenna configurations with half-power beamwidths (HPBWs) of 13° (21 dBi), 9° (27 dBi), 4° (34 dBi), and 1.3° (40 dBi), using link availability to assess the gain-beamwidth trade-off under mobility.

The Hierarchical Search and IMU-Assist algorithms are ineffective across all antenna types under combined motion (Fig. 8). The rotational component causes Hierarchical Search to steer the AP away from the correct LoS, keeping availability below 9% regardless of beamwidth. IMU-Assist peaks at 43% with the 13° antenna but falls below 10% above 2 m/s^2 and 80 deg/s^2 for all configurations.

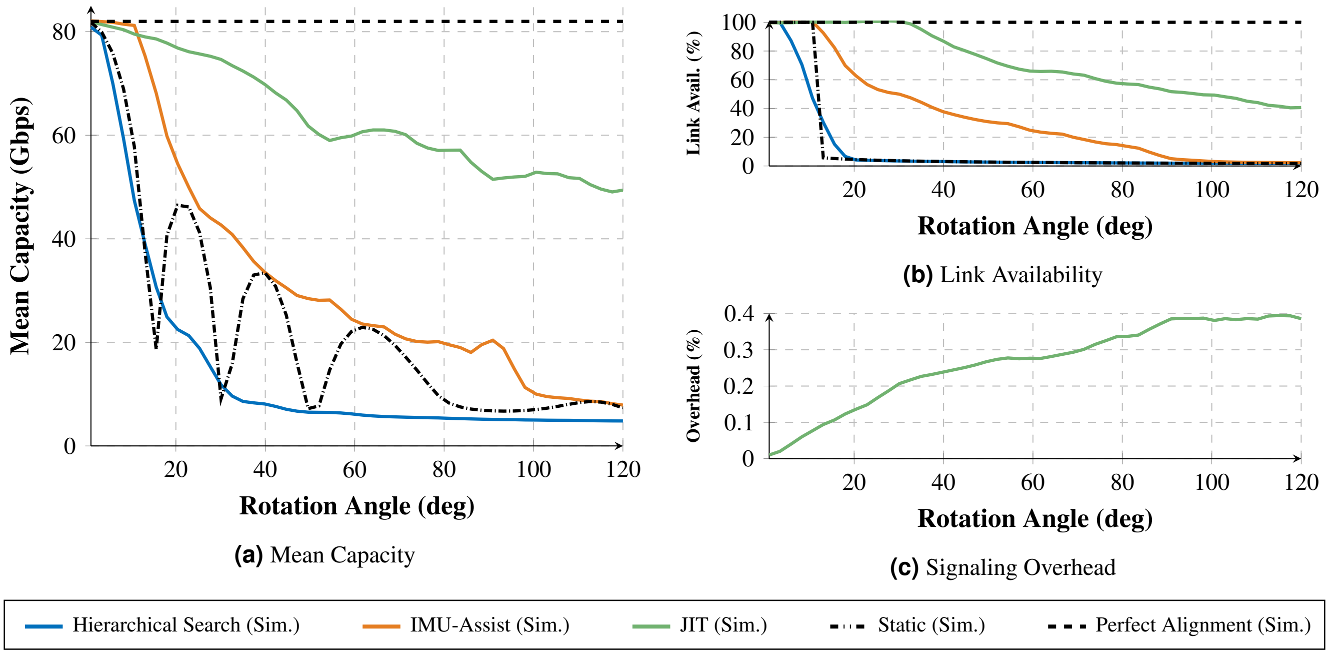


Figure 7. Effect of total angular displacement (1°–120°) at fixed rotational acceleration (120 deg/s^2), Ideal Mechanical. (c) Overhead plateaus at ∼0.39% above 90° as realignment frequency saturates.

JIT reveals a trade-off between gain and beamwidth. With the 13° antenna (Fig. 8a), the wide beam relaxes pointing requirements, but the lower gain (21 dBi) limits the SNR margin. JIT achieves >80% availability only between 0.6 m/s^2 to 1.7 m/s^2 before transient pointing errors push SNR below the 18.8 dB threshold. The 9° antenna (Fig. 8b) provides the best balance: its higher gain (27 dBi) buffers against misalignment, extending the >80% availability window to 0.55 m/s^2 to 2.85 m/s^2 and sustaining above 63% even at maximum acceleration.

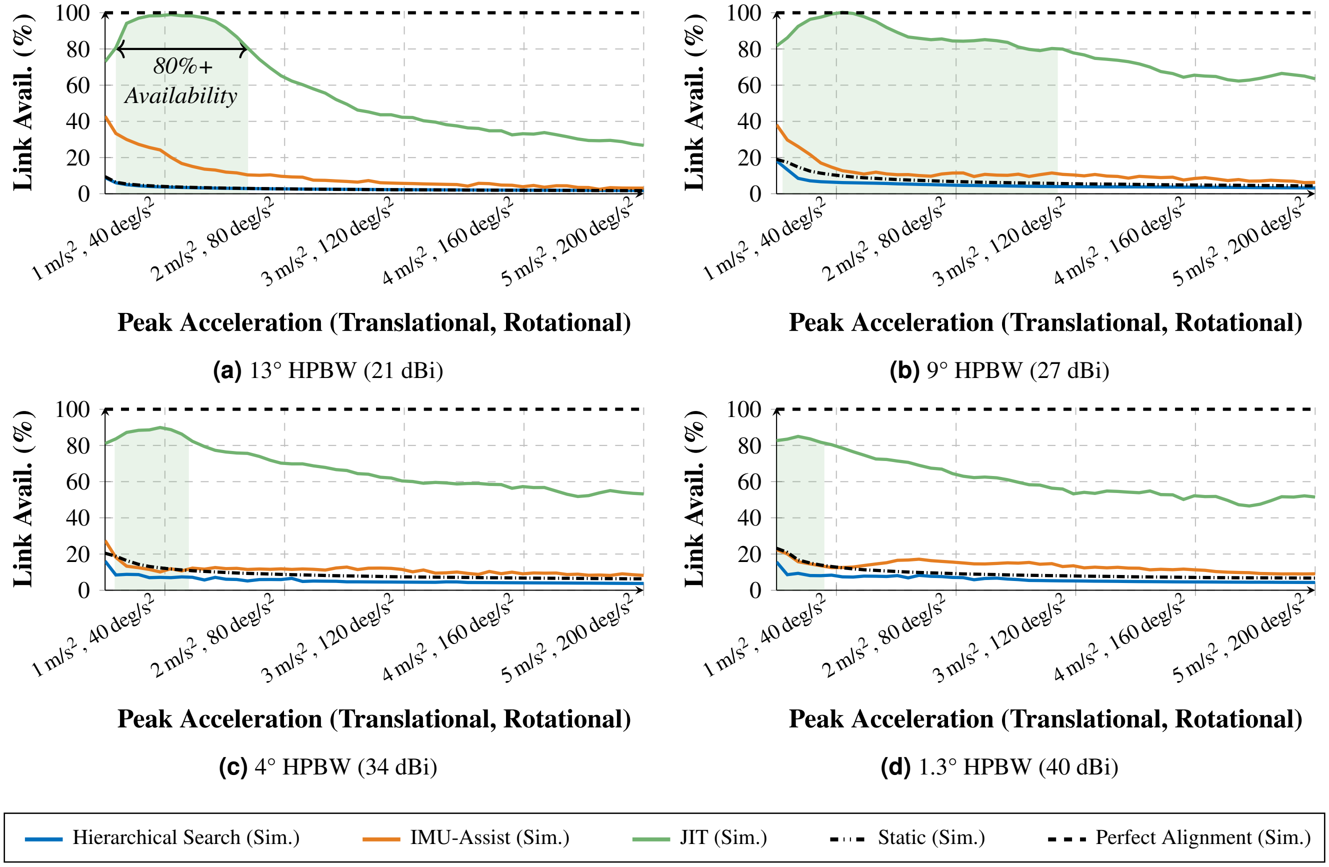
With the 4° antenna (Fig. 8c), the >80% window narrows to 0.58–1.2 m/s^2 as the tighter beam demands higher pointing precision than the algorithm's reaction time can guarantee. The 1.3° antenna (Fig. 8d) restricts this further to 0.5–0.9 m/s^2, with availability dropping below 80% before reaching 1 m/s^2. Despite their higher gain (34–40 dBi), the angular displacement during the system's reaction latency exceeds the beamwidth, negating the link budget advantage. These results show that each mobility profile imposes a minimum usable beamwidth, and that selecting the right antenna requires jointly considering gain, control latency, and expected user motion.

Discussion

The above results characterize mobile sub-THz link performance as a function of link geometry, motion type and severity, and beam steering hardware, comparing four beam realignment strategies of increasing sophistication: a static baseline, a reactive Hierarchical Search, a reactive IMU-Assisted approach in line with prior work at lower frequencies[57,58], and the proposed hybrid proactive+reactive JIT beam realignment. *To our knowledge, this constitutes one of the first comprehensive studies of a hybrid proactive and reactive beam realignment technique for mobile sub-THz communications validated through HIL measurements with sub-THz RF hardware.* The results, corroborated by both simulation and experimental measurements, lead to several principal conclusions.

The **first** key conclusion is that building a reliable and high-performance truly mobile sub-THz communication system demands an upgrade from the state-of-the-art *reactive* beam management procedures to novel *proactive* and hybrid *proactive+reactive* beam management solutions. Specifically, the results in Fig. 5 illustrate that the link availability for the reactive Hierarchical Search solution never rises above 40% for moderate-to-low mobility levels. Hence, such an unreliable communication link cannot be utilized in practice for any latency-sensitive or reliability-sensitive (e.g., XR) services envisioned for mobile sub-THz communications. In contrast, the proposed JIT solution demonstrates superior performance here, achieving up to $90+\%$ (sometimes, up to 100%) link availability in the same conditions.

The **second** key conclusion from the present study is that the sole presence of an IMU at a mobile node and the use of IMU data just in the beam realignment procedure (as discussed in several prior works for lower frequencies, including microwave and mmWave) becomes insufficient for mobile sub-THz. Although the results in Figs. 2–7 show that IMU-Assist



(a) 13° HPBW (21 dBi)

(b) 9° HPBW (27 dBi)

(c) 4° HPBW (34 dBi)

(d) 1.3° HPBW (40 dBi)

Figure 8. Combined mobility (1.5 m translation + 90° rotation, escalating acceleration) for four antenna configurations. Dashed line marks 80% availability. The 9° HPBW antenna yields the widest operational window, illustrating the gain–beamwidth trade-off under reaction-time constraints.

generally outperforms Hierarchical Search in both performance and reliability, its absolute link availability remains notably lower than what the proposed JIT approach achieves. This contrast is particularly visible in Fig. 8 presenting the link availability for different acceleration levels and different antenna configurations. JIT achieves over 80% link availability across a wide operating range, even with the narrowest 1.3° HPBW antenna. By contrast, the peak availability for reactive-only IMU-Assist reaches only ∼40%, and its average falls 3–4 times lower, notably under 20%. Hence, we may conclude that for mobile sub-THz communications featuring narrow-beam antennas, not every IMU-assisted beam realignment approach is sufficient. Our results suggest that proactive approaches – those that utilize IMU data to continuously improve link quality, rather than only upon link interruption – are needed to deliver the required performance.

Notably, as further confirmed by our study, while the JIT beam alignment approach introduces a signaling overhead, the penalty imposed by this overhead is minimal compared to the gains in capacity and reliability. Our results further indicate that the signaling overhead scales linearly with the rotation angle, increasing from near 0 % to a maximum of only 0.39 % of the total link capacity (Fig. 7c). This represents a negligible penalty relative to the significant improvements in mean capacity and link availability. In summary, the additional control data required for JIT may be sent with negligibly low signaling overhead, thus not decreasing the capacity of the sub-THz link during the data exchange phase.

The **third** key conclusion from our performed study is that the choice of the particular beam steering hardware platform (e.g., different mechanical beam steering vs. electronic beam steering solutions) is equally if not more important than the choice of the particular beam realignment algorithms and protocols. The mean capacity results in Fig. 2 differ substantially across beam steering methods: constrained mechanical, ideal mechanical, and electronic. Although the proposed JIT protocol can minimize the impact of hardware constraints, particularly in latency-bound scenarios, the underlying steering hardware remains a dominant performance factor. Nevertheless, JIT effectively mitigates these latency limitations by decoupling failure prevention from failure recovery. Specifically, using a just-in-time trigger based on sensor fusion, the system maintains low capacity variance, reflected in a CoV of approximately 0.1 as observed in the translational mobility analysis (Fig. 3b). Low variance here is critical for transport-layer protocols (e.g., TCP) that are sensitive to large variations in packet loss and jitter[59].

Our **fourth** major conclusion from the performed study is that the design of sub-THz nodes that are simultaneously high-performance, robust, and truly mobile requires a shift in the antenna selection approach. While traditional designs for stationary sub-THz connectivity systems prioritize maximizing gain to overcome path loss and achieve the highest peak SNR values, our results confirm that mobility introduces a competing constraint: narrower beams demand faster and more precise steering, and the system's control latency limits how narrow a beam can be used reliably under motion. Although higher gain antennas improve SNR, their narrower beamwidths reduce the system's tolerance for pointing errors and increase sensitivity to control latency. In our study, the 13° antenna suffered from insufficient gain to maintain the link during transient slewing events, while the 4° and 1.3° antennas failed because the angular displacement during the reaction time exceeded their beamwidths. For our particular configuration, we identified an optimal operating region—represented by the 9° antenna—that balanced gain margin with alignment tolerance. Although these specific parameters are unique to our experimental hardware and evaluation scenario, the broader implication is that antenna selection must account for the platform's specific mobility profile and control latency. Here, the new MobileTHz platform developed in this work also serves as a useful solution to characterize this antenna-related trade-off. It particularly allows communication engineers to identify the optimal antenna parameters required to support different classes of movement for future truly mobile sub-THz communication systems.

Finally, while the presented results and analysis indicate very promising performance improvements brought by implementing JIT-type beam management in future mobile sub-THz networks, several avenues remain for extending JIT toward an even more intelligent and robust mechanism in *future work*. First, the current hard-coded rule-based realignment could be better tailored to specific environments using learning-based decision making[35], whether AI-driven or based on advanced heuristics. Second, the LoS assumption can be further relaxed. Specifically, the present evaluation does not consider transient blockage events (e.g., a hand or body obstructing the beam). However, by design, the JIT framework offers a natural path toward blockage mitigation as well. If control frames are transmitted periodically during steady-state operation (leveraging also its extremely low signaling overhead discussed above) rather than only at the EoC boundary, the AP would receive continuous kinematic updates. A sudden RF power drop or packet loss *without* a corresponding change in the UE's reported motion state would indicate an external obstruction rather than misalignment, enabling the AP to suppress unnecessary beam searches and instead await blockage clearance, leverage multi-connectivity, initiate handover, or employ another blockage mitigation technique. Hence, in principle, the approach discussed in this article can be developed further to also cover blockage mitigation caused by mobile obstacles, another pressing challenge for sub-THz connectivity besides the beam alignment. Third, the present evaluation study is focused on a single AP–UE link. Meanwhile, expanding the proposed approach toward a multi-user deployment, the per-user kinematic data carried by JIT could not only facilitate beam alignment, but also enable mobility-aware beam scheduling and more favorable scaling of aggregate steering overhead compared to the state-of-the-art reactive methods.

Summarizing, while the presented first-order evaluation results already show substantial performance improvement, when utilizing the hybrid proactive+reactive JIT beam management approach (up to 1–2 orders of magnitude improvement in certain metrics and setups), our proposed approach can also be developed further both (i) in terms of its efficiency in mitigating beam misalignment events, and (ii) by expanding the original sub-THz beam realignment, e.g., to also facilitate blockage mitigation or more efficient radio resource management in future mobile sub-THz wireless communication systems beyond 6G.

Methods

The MobileTHz platform comprises two main components: a modular C++ simulation engine (that we made publicly available[60]) and an IMU-integrated HIL testbed further detailed in the "The Experimental Platform" subsection below. The hardware platform substantially extends the version reported in our earlier conference paper[61]. Together, these tools enable comprehensive evaluation of beam management algorithms across diverse mobile sub-THz scenarios. All algorithms are evaluated across three steering models of increasing actuation speed (see Table 1), spanning from (i) laboratory-level equipment-constrained mechanical hardware through (ii) idealized mechanical beam steering up to (iii) electronic phased-array beam steering.

Table 1. Steering-model configurations.

	Constrained Mechanical	Ideal Mechanical	Electronic
Mechanism	2-Stage rotary	2-Stage rotary	Phased array
$\omega_{\max}$	$60\,^\circ/\mathrm{s}$	$720\,^\circ/\mathrm{s}$	—
$\alpha_{\max}$	$60\,^\circ/\mathrm{s}^2$	$3600\,^\circ/\mathrm{s}^2$	—
Steer time	—	—	5 µs
Resolution	0.0005°	0.001°	θ_{HPBW}

To assess the proposed JIT solution, we also implemented four alternative algorithms into the MobileTHz platform. The two primary comparison algorithms are described below in detail: (i) *"Hierarchical Search"* – a state-of-the-art sequential beam realignment approach not leveraging any IMU assistance; and (ii) *"IMU-Assist"* – an advanced solution leveraging IMU data to faster and better re-align the misaligned sub-THz beams but not utilizing the proactive mechanisms or control frames employed by JIT. The other two algorithms include the reference lower bound *"Static"* (no realignment) approach and the reference upper bound *"Perfect Alignment"* setup, employing an Oracle to tell both the AP and UE to where to point their beams best at each moment of time.

Hierarchical Search

Hierarchical Search operates independently and identically at both the AP and UE. Since the procedure is triggered by link failure, no communication between the nodes is possible during recovery; each node executes its own search using only local power measurements.

The procedure is initiated when a short-term Exponentially Weighted Moving Average (EWMA) of received power drops below a failure threshold $P_{\text{fail}} = P_{\text{ref}} - \Gamma_{\text{fail}}$, where P_{ref} is the reference power (the last successfully received power level before failure) and Γ_{fail} is the failure margin (parameters in Table 2). Upon detection, each node independently executes a discrete spiral scan outward from its reference beam direction (Fig. 9), adapted from optical alignment systems[62]. The step size for successive rings, HS_k, grows exponentially from an initial value proportional to the antenna's HPBW:

$$HS_k = HS_0 \cdot \gamma^k, \tag{1}$$

where k is the ring index, $HS_0 = \theta_{\text{HPBW}}/2$ is the initial step size (half the antenna beamwidth), and $\gamma = 1.25$ is the growth factor. The scan terminates when the received power for a given beam configuration exceeds a time-dependent success threshold $P_{\text{succ}}(\Delta t)$:

$$P_{\text{succ}}(\Delta t) = P_{\text{ref}} - \Gamma(\Delta t) = P_{\text{ref}} - (\Gamma_0 + R \cdot \Delta t), \tag{2}$$

where Γ_0 is the initial margin, R is the margin increase rate, and Δt is the elapsed time since failure detection. The relaxing margin accounts for the growing positional uncertainty: as more time elapses, the node accepts weaker beam configurations to avoid indefinite searching. Once a beam configuration exceeding $P_{\text{succ}}(\Delta t)$ is found, the node holds that beam direction and resumes data exchange.

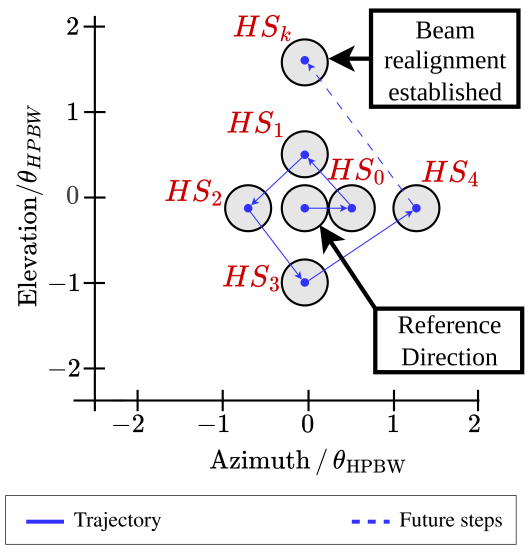


Figure 9. Hierarchical Search spiral trajectory expanding outward from the reference beam direction with step sizes $HS_0, HS_1, \ldots, HS_k$. Circles show the evaluation order; the scan ends when received power exceeds $P_{\text{succ}}(\Delta t)$.

IMU-Assisted Reactive Search

The IMU-Assisted algorithm uses the same power-drop trigger as Hierarchical Search but replaces the undirected spiral beam steering trajectory with a refined sensor-directed steering trajectory (Fig. 10). This procedure is built upon a principle of Kalman-filter-based predictive tracking[38,39] but operates in a triggered rather than continuous mode. Upon failure detection, the UE integrates accelerometer data into a displacement vector $\Delta\mathbf{p}(t)$ and fuses gyroscope data for orientation change, predicting the new beam pointing direction via:

$$\theta^{\star}_{\text{az}}(t) = \theta_{\text{az},0} + \arctan\left(\frac{\Delta p_y}{\hat{d}_{\text{link}}}\right) + \Delta\psi(t) \tag{3}$$

$$\theta^{\star}_{\text{el}}(t) = \theta_{\text{el},0} + \arctan\left(\frac{\Delta p_z}{\hat{d}_{\text{link}}}\right) + \Delta\phi(t) \tag{4}$$

where $\theta_{\text{az},0}$ and $\theta_{\text{el},0}$ are the last known-good angles; Δp_y and Δp_z are the horizontal and vertical components of displacement perpendicular to the LoS; d_{link} is the estimated AP–UE link distance from the last known-good reference time before the link failure; and $\Delta\psi(t)$ and $\Delta\phi(t)$ are the changes in yaw and pitch, respectively. The UE then steers directly to this predicted angle. However, because IMU-based positioning relies on double-integrating noisy accelerometer readings, small measurement errors accumulate over time, causing the estimated position and orientation to progressively diverge from their true values (a phenomenon known as integration drift). To compensate for this residual sensor error,

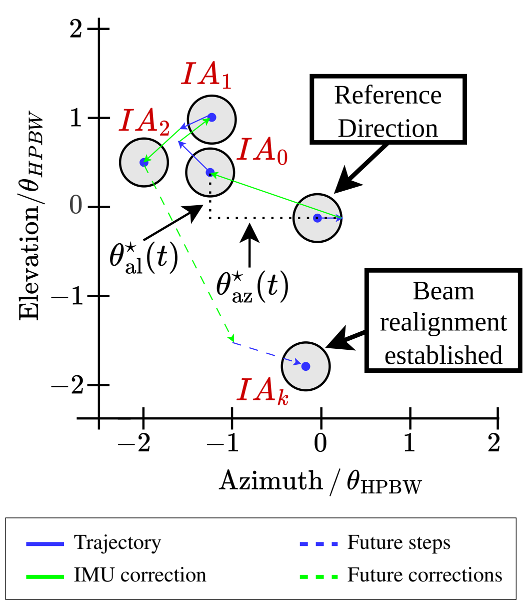


Figure 10. IMU-Assisted search trajectory. IMU corrections (green) shift the search center to the sensor-predicted LoS direction before each spiral step. The local refinement spiral (blue) scans around the corrected center.

the algorithm follows the predicted steer with a localized refinement phase identical in purpose to the one used by Hierarchical Search but with a tighter step size:

$$IA_k = \beta \cdot \theta_{\text{HPBW}} + \Delta\theta^{\star}_{\text{az}}(t) + \Delta\theta^{\star}_{\text{el}}(t), \quad (\beta = 0.1) \tag{5}$$

Unlike the expanding step size HS_k used in the Hierarchical Search's Search Phase, the base refinement step $\beta \cdot \theta_{\text{HPBW}}$ is fixed and deliberately small (one-tenth of the beamwidth), while the IMU-predicted azimuth and elevation corrections adapt IA_k over time as the sensor estimates evolve. The node scans a tight spiral around the IMU-predicted direction using this step size until an alignment is made (see Fig. 10).

JIT Beam Realignment

While the reactive IMU-Assist algorithm discussed above aims to minimize the beam realignment time once the link failed, our proposed proactive+reactive JIT strives to also prevent the link failure from happening (by realigning the AP and UE beams continuously while the link is still in place). At the same time, not to compromise the actual data exchange and link performance, JIT strives not to produce control signaling overheads and beam re-steering actions unless it is necessary to prevent the link failure. This goal is achieved by designing the system that only triggers beam realignment when the combination of motion speed and link geometry threatens to place the UE outside the AP's beam before the next control cycle.

For this purpose, since any practical IMU sensor features noise in its data, a stationary Zero Velocity Update (ZUPT) phase is performed at initialization to calibrate the accelerometer bias $\mathbf{b}_a$ and gyroscope bias $\mathbf{b}_\omega$. During this phase, the UE remains stationary for a brief period, and the mean of the raw accelerometer and gyroscope readings is computed. Since the true acceleration and angular velocity are both zero during this interval, the computed means directly estimate the sensor biases. These bias estimates are then subtracted from all subsequent IMU readings to obtain corrected acceleration $\mathbf{a}_{\text{corr}}[n] = \tilde{\mathbf{a}}[n] - \mathbf{b}_a$ and angular velocity $\omega_{\text{corr}}[n] = \tilde{\omega}[n] - \mathbf{b}_\omega$, enabling the system to distinguish link-critical motion from sensor noise. Utilizing the corrected IMU data, the UE then continuously computes a new metric, which we refer to as "Link Deviation Velocity", $\dot{\delta}$. This metric unifies rotational and LoS-perpendicular translational UE motion into a single angular rate of deviation:

$$\dot{\delta} = \omega_{\text{UE}} + \frac{\mathbf{v}_\perp}{d_{\text{link}}}, \tag{6}$$

where ω_{UE} is the UE's angular velocity, $\mathbf{v}_\perp$ is its translational velocity perpendicular to the LoS, and d_{link} is the instantaneous link distance. This metric hence serves as measure of "how fast the UE movements are approaching the AP beam boundaries".

Finally, a control frame is triggered when the projected angular displacement over the system's reaction time T_{react} exceeds a fraction η of the antenna's HPBW:

$$\|\dot{\delta}\| \cdot T_{\text{react}} > \eta \cdot \theta_{\text{HPBW}} \tag{7}$$

The guard fraction η trades off link robustness against control overhead in the JIT algorithm: a lower η triggers control frames earlier, providing more margin against sudden motion at the cost of higher signaling frequency, while a higher η reduces overhead but leaves less time to react before the beam exits the main lobe. We find $\eta = 0.36$ effective across our evaluated scenarios.

Upon triggering, the UE transmits a control frame containing its integrated displacement vector and a buffer of recent IMU samples. The AP applies an EWMA to the buffered samples to derive a filtered acceleration trend $\bar{\mathbf{a}}_{\text{predict}}$, extrapolating the UE's position at steering completion. By steering to this predicted future location rather than the last known position, the system maintains SNR continuity despite rapid motion.

Each control frame follows the IEEE 802.15.3d THz-SC structure[20], using a short Preamble Acquisition Preamble (PAP) of 1.89 µs and the minimum 2048-byte frame payload. With worst-case BPSK modulation and 11/15 LDPC coding, each control frame occupies approximately $T_{\text{frame}} \approx 5\,\mu\text{s}$. The end-to-end reaction latency comprises the control frame transmission time (T_{frame}) and the AP's signal acquisition and processing delay ($T_{\text{proc}} = 10\,\mu\text{s}$), giving a total signaling latency of $T_{\text{react}} = T_{\text{frame}} + T_{\text{proc}} \approx 15\,\mu\text{s}$ before the AP can issue a steering command. Reliable decoding at the AP requires a minimum receive SNR of 9.59 dB, corresponding to BPSK with 11/15 LDPC coding at a target BER $= 10^{-3}$. This threshold determines the maximum range at which control frame remains decodable: if the instantaneous SNR falls below 9.59 dB, the AP cannot decode the control frame and the system falls back to reactive recovery: the UE continues to steer using its local IMU data, while the AP, lacking motion awareness, initiates a Hierarchical Search to restore alignment.

The MobileTHz Simulation Platform

The MobileTHz platform discretizes time into fixed-duration slots of length $\Delta t_{\text{sim}} = 1\,\mu\text{s}$, enabling deterministic, slot-by-slot evaluation of node kinematics, RF channel state, hardware behavior, and algorithm decisions. Simulation parameters, listed in Table 2, mirror the experimental hardware properties to ensure close correspondence between simulated and measured results.

Table 2. Shared system parameters for the channel model, transceiver, IMU sensor, and beam-management algorithms. All values are common to the three steering configurations.

Parameter	Value	Parameter	Value
Channel		***IMU Sensor***	
Carrier Frequency (f_c)	130 GHz	Gyro. Noise Density	$0.00175\,\mathrm{rad/s/\sqrt{Hz}}$
Bandwidth (B)	8 GHz	Accel. Noise Density	$0.0294\,\mathrm{m/s^2/\sqrt{Hz}}$
System Temperature (T)	290 K	Initial Bias Std. Dev.	0.01 (gyro/accel)
Thermal Noise (k_BTB)	−75.0 dBm	Sampling Period	10 ms
Transceiver		***Algorithm Parameters***	
Tx Power (P_{tx})	13 dBm	Failure Margin (Γ_{fail})	3 dB rel. P_{ref}
Rx Noise Factor (F)	14.83 (11.7 dB)	EWMA Coeff. (α_{short})	0.90
Rx Processing Delay (T_{proc})	10 µs	Spiral Growth Factor (γ)	1.25
		Initial Success Margin (Γ_0)	0.25 dB
		Margin Increase Rate (R)	$0.5 \cdot G_{\max}$ dB/s

A mobile scenario consists of a set of mobile nodes, $\mathscr{M} = \{m_1, m_2, \dots\}$, whose trajectories are specified by a sequence of keyframes, $\{(\mathbf{p}_k, \mathbf{q}_k)\}_{k=0}^{K-1}$. Each keyframe defines the position $\mathbf{p}_k \in \mathbb{R}^3$ and orientation (as a unit quaternion) $\mathbf{q}_k \in \mathbb{S}^3$ at a specific time slot n_k. To create smooth and realistic translational paths between keyframes, the platform interpolates position using Perlin's smootherstep function[63], $f(\tau)$:

$$f(\tau) = 6\tau^5 - 15\tau^4 + 10\tau^3, \tag{8}$$

where $\tau = (t - t_k)/(t_{k+1} - t_k) \in [0,1]$ is the normalized time. This function's C^2 continuity guarantees zero velocity, acceleration, and jerk at keyframe boundaries, producing physically plausible motion. The resulting maximum acceleration for a displacement D over duration T is:

$$a_{\max} = K \cdot \frac{D}{T^2}, \tag{9}$$

where $K \approx 5.7735$ is derived from the peak of the second derivative of Eq. (8). Orientation, in turn, is interpolated via Spherical Linear Interpolation (SLERP):

$$\text{SLERP}(\mathbf{q}_k, \mathbf{q}_{k+1}, \tau) = \frac{\sin((1-\tau)\Theta)}{\sin\Theta}\mathbf{q}_k + \frac{\sin(\tau\Theta)}{\sin\Theta}\mathbf{q}_{k+1}, \tag{10}$$

where $\Theta = \arccos(\mathbf{q}_k \cdot \mathbf{q}_{k+1})$. Translational and angular velocities and accelerations are obtained via central finite differences of position and orientation, respectively.

Given the updated node kinematics, the engine evaluates the RF link state. Free-space path loss (FSPL), L_p, in dB is:

$$L_p = 20\log_{10}(d_{\text{link}}) + 20\log_{10}(f_c) - 147.55, \tag{11}$$

where $d_{\text{link}} = ||\mathbf{p}_{rx} - \mathbf{p}_{tx}||$ is the instantaneous LoS distance in meters and f_c is the carrier frequency in Hz. Antenna directivity is evaluated from the node's effective orientation $\mathbf{q}_{eff} = \mathbf{q}_{body} \otimes \mathbf{q}_{rotary}$, which transforms the LoS vector into the antenna's local frame to obtain the off-axis angle. The directivity follows the E-plane pattern of a WR-6.5 conical horn, whose gain $G(\theta)$ is given by:

$$G(\theta) = \frac{1+\cos(\theta)}{2}\sqrt{\frac{(C(r_2)-C(r_1))^2 + (S(r_2)-S(r_1))^2}{4(C(r_0)^2 + S(r_0)^2)}}, \tag{12}$$

where $C(x)$ and $S(x)$ are the Fresnel cosine and sine integrals with arguments:

$$r_0 = \frac{b_{\text{wg}}}{2}\sqrt{\frac{\beta_g}{\pi R_h}}, \quad r_{1,2} = \sqrt{\frac{\beta_g}{\pi R_h}}\left(\mp\frac{B_E}{2} - \frac{R_h\beta_g B_E}{2}\sin\theta\right), \beta_g = \frac{2\pi}{\lambda}\sqrt{1-\left(\frac{\lambda}{2a_{\text{wg}}}\right)^2} \tag{13}$$

Here $R_h = 25$ mm is the horn slant length, $B_E = 3.59$ mm is the effective E-plane aperture, $b_{\text{wg}} = 0.826$ mm and $a_{\text{wg}} = 1.651$ mm are the waveguide height and width, respectively. Then, the maximum gain is defined as $G_{\max} = \max_\theta G(\theta)$. The received power (in dBm) then combines transmit power, directional gains, and path loss:

$$P_{rx} = P_{tx} + G_{tx}(\theta_{tx}) + G_{rx}(\theta_{rx}) - L_p, \tag{14}$$

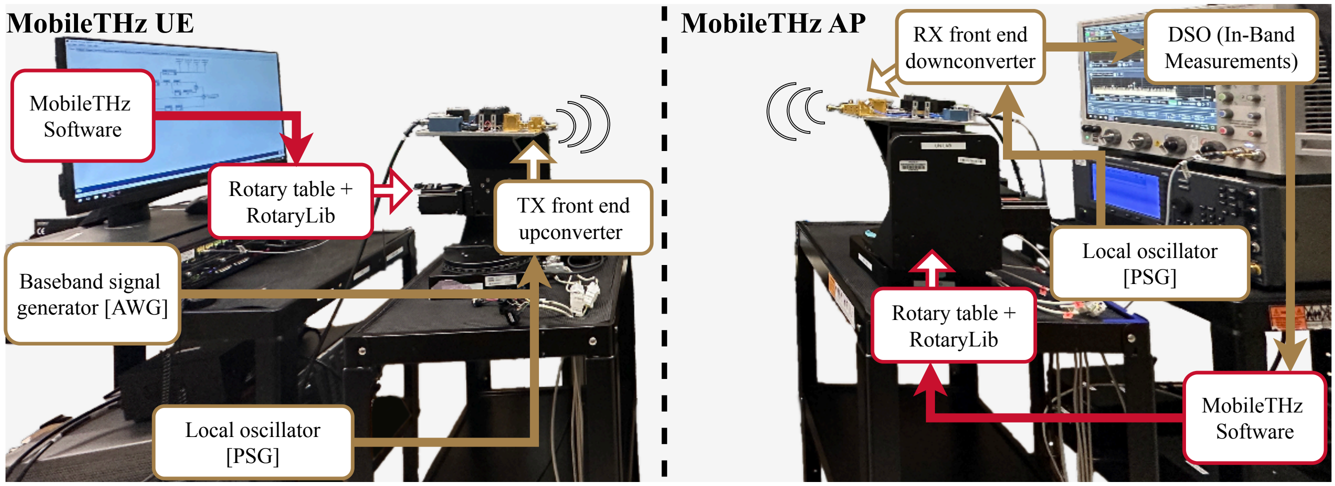


Figure 11. MobileTHz HIL testbed. The UE (left) and AP (right) each pair a sub-THz RF front-end with a dual-axis rotary stage. A 9-axis IMU on the UE supplies inertial data for the IMU-Assisted and JIT algorithms.

where P_{tx} is the transmit power and θ_{tx}, θ_{rx} are the off-axis angles at the transmitter and receiver, respectively.

The simulation then applies a hardware abstraction layer that models the constraints of physical components. A steerable antenna's orientation is controlled by a dual-axis mechanical stage, which adjusts the $\mathbf{q}_{rotary}$ component of the effective orientation. For any instructed angular displacement $\Delta\theta$, the model uses a trapezoidal velocity profile:

$$a(t) = \begin{cases} +\alpha_{\max} & \text{Phase I: Acceleration} \\ 0 & \text{Phase II: Constant velocity} \\ -\alpha_{\max} & \text{Phase III: Deceleration} \end{cases}, \tag{15}$$

where the duration of each phase is determined by $\Delta\theta$, the stage's maximum angular velocity, ω_{max}, and the angular acceleration constraints, α_{max}.

Beyond mechanical steering, the hardware abstraction layer captures two additional sources of latency and noise. A configurable processing delay T_{proc} models signal acquisition and digital conversion latency, holding the power reading constant at the value sampled at the start of the previous processing interval. The simulated IMU provides gyroscope (ω), accelerometer ($\mathbf{f}$), and magnetometer data, with realistic outputs generated by corrupting ground-truth kinematics with bias and noise. Denoting the discrete time-slot index by n, the simulated accelerometer output $\tilde{\mathbf{a}}[n]$ is:

$$\tilde{\mathbf{a}}[n] = \mathbf{f}_{ideal}[n] + \mathbf{b}_a[n] + \mathbf{n}_a[n]. \tag{16}$$

The bias $\mathbf{b}_a[n]$ evolves as a random walk: $\mathbf{b}_a[n] = \mathbf{b}_a[n-1] + \mathbf{w}_{b,a}[n]$, where the update term $\mathbf{w}_{b,a} \sim \mathcal{N}(0, \sigma_{a,b}^2 \Delta t_{\text{sim}} \cdot \mathbf{I}_3)$ and the white noise is $\mathbf{n}_a[n] \sim \mathcal{N}(0, (\sigma_{a,n}^2/\Delta t_{\text{sim}}) \cdot \mathbf{I}_3)$. A similar noise model is applied to the gyroscope readings. The noise density parameters $\sigma_{a,n}$ and bias random walk parameter $\sigma_{a,b}$ correspond to the Accelerometer Noise Density and Initial Bias values in Table 2, respectively.

This sequence of kinematics update, channel evaluation, hardware emulation, and algorithm decision repeats at every 1 µs-long time slot, producing the continuous per-slot feedback loop that drives all simulated results in this article.

The Experimental Platform

A similar per-slot loop calculation procedure works in the HIL mode, employing the hardware sub-THz platform. However, the simulated RF and sensor inputs discussed in the previous subsection get replaced by actual physical measurements. Specifically, the received power data comes from the RF hardware, orientation data gets fetched from rotary-stage encoders, and inertial data from an on-device IMU. The complete testbed is illustrated in Fig. 11.

The RF chain operates in the WR-6.5 waveguide band (110–170 GHz). On the transmit side, a Keysight E8257D PSG and M8196A AWG feed a VDI MixAMC upconverter; on the receive side, a VDI MixAMC downconverter and Keysight DSOZ632A oscilloscope capture in-band power. Each front-end is mounted on an IntelLiDrives dual-axis rotary table equipped with a VDI WR-6.5 conical horn antenna (21 dBi, 13° HPBW). A WitMotion WT901BLECL 9-axis IMU, co-located with the UE front-end, supplies the inertial data used by the IMU-Assisted and JIT algorithms.

A custom in-house-developed C++ library (RotaryLib) controls the rotary table movements by sending commands via serial connection. Each commanded motion follows the same trapezoidal acceleration-cruise-deceleration profile assumed by the simulator, ensuring that physical stage behavior matches the simulated steering model. Power measurements, encoder feedback, and IMU samples are streamed back to the MobileTHz engine, allowing the same beam management algorithm code to run on physical data without modification.

Evaluation Metrics

We evaluate the proposed beam management algorithms across four Key Performance Indicators (KPIs) derived from the RF and mobility parameters summarized in Table 2 and Table 1. They complement each other, allowing us to evaluate the performance and reliability of various beam realignment strategies from different perspectives.

Mean Link Capacity serves as the primary performance metric, calculated as the time-average of instantaneous Shannon capacity $C(t)$ over the duration of the scenario T:

$$\bar{C} = \frac{1}{T}\int_0^T B\log_2(1+\mathrm{SNR}(t))\,dt \tag{17}$$

where B represents the channel bandwidth. However, as the link is not always available due to beam misalignment, the calculated mean capacity also obscures the impact of beam misalignment and outages on higher-layer protocols (e.g., TCP).

The second metric of interest in our study is the *Capacity Coefficient of Variation (CoV)* used to quantify the capacity variance over time. To compute the CoV, we normalize the standard deviation of capacity (σ_C) by its mean (μ_C):

$$\mathrm{CoV} = \frac{\sigma_C}{\mu_C} \tag{18}$$

High CoV values indicate erratic tracking and jitter, whereas lower values approaching zero indicate a more stable and reliable link.

To further assess the operational feasibility of the link for actual data transmission, we then evaluate *Link Availability (A_{link}*. This metric measures the fraction of time the link budget is sufficient to support high-order modulation schemes, specifically where the $\mathrm{SNR}(t)$ exceeds a critical threshold $\mathrm{SNR}_{\mathrm{th}}$:

$$A_{\mathrm{link}} = \frac{1}{T}\int_0^T \mathbb{1}(\mathrm{SNR}(t) \geq \mathrm{SNR}_{\mathrm{th}})\,dt \times 100\% \tag{19}$$

The threshold $\mathrm{SNR}_{\mathrm{th}} = 18.8\,\mathrm{dB}$ corresponds to 64-QAM at a FEC BER $= 10^{-3}$, derived from the standard M-QAM approximation[64]:

$$\mathrm{SNR}_{\mathrm{th}} \approx \frac{M-1}{3}\left[Q^{-1}\left(\frac{\mathrm{BER}_{\mathrm{target}}\cdot\log_2 M}{4}\right)\right]^2 \tag{20}$$

which evaluates to 18.8 dB.

Last but not least, we also quantify the bandwidth cost (overheads) of JIT's additional control frames via Signaling Overhead (O_{sig}). The rationale here is to provide a fair evaluation, since the link cannot carry any user data during each control frame transmission. Hence, while the JIT-imposed additional link robustness contributes to the channel capacity and availability, this comes at a cost of additional signaling overhead that has to be carefully estimated.

The foregone capacity over N_{pp} transmissions is:

$$L_{\mathrm{ctrl}} = \sum_{i=1}^{N_{\mathrm{pp}}} C(t_i)\cdot T_{\mathrm{frame}}, \tag{21}$$

where $C(t_i)$ is the instantaneous capacity at the i-th transmission and T_{frame} is the control frame duration derived in the JIT Algorithm subsection. Signaling overhead is then:

$$O_{\mathrm{sig}} = \frac{L_{\mathrm{ctrl}}}{\int_0^T C(t)\,dt} \times 100\% \tag{22}$$

While the considered beam realignment strategies can, in principle, be evaluated in our platform across multiple other metrics, the selected set of four diverse KPIs already present a comprehensive view on the key trade-offs.

Statistics and Reproducibility

All comparisons between algorithms are based on deterministic metrics (Mean Capacity, Capacity CoV, Link Availability, Signaling Overhead) computed over the full scenario duration for each configuration. All percentage improvements reported between algorithms (e.g., "JIT achieves up to an 88% improvement over IMU-Assist") are computed directly from the respective mean capacity or link availability values at matched operating points. These comparisons reflect differences in algorithmic design evaluated under identical controlled conditions.

Each simulation data point averages $n = 5{,}000$ independent Monte Carlo runs. The source of randomness depends on the scenario: in translational cases, the UE movement direction is drawn uniformly from $[0, 2\pi)$ with no rotation; in rotational cases, the rotation sense and axis are randomized with no translation; in combined-motion cases, both direction and rotation are randomized; and in link-geometry sweeps, the motion profile is fixed while the geometric parameter (separation distance or angular displacement) is varied deterministically. All other parameters (carrier frequency, transmit power, antenna configuration, steering model) are kept constant within each sweep as specified in Tables 1 and 2.

Each experimental (HIL) data point averages $n = 10$ independent measurement trials. Experiments were constrained to the azimuth plane by the weight of the RF equipment. Agreement between simulation and experiment is quantified using MAPE and NRMSE, computed as:

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{\bar{C}_{\mathrm{sim},i} - \bar{C}_{\mathrm{exp},i}}{\bar{C}_{\mathrm{sim},i}}\right| \times 100\% \tag{23}$$

$$\mathrm{NRMSE} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(\bar{C}_{\mathrm{sim},i} - \bar{C}_{\mathrm{exp},i})^2}}{\bar{C}_{\mathrm{sim,max}} - \bar{C}_{\mathrm{sim,min}}} \times 100\% \tag{24}$$

where $\bar{C}_{\mathrm{sim},i}$ and $\bar{C}_{\mathrm{exp},i}$ are the simulated and experimental mean capacities at the i-th acceleration point, N is the number of acceleration points in the sweep, and $\bar{C}_{\mathrm{sim,max}}$ and $\bar{C}_{\mathrm{sim,min}}$ are the maximum and minimum simulated mean capacities across the sweep.

Data Availability

All data generated or analysed during this study are included in this published article and its supplementary information files. Additional data can be made available to readers upon reasonable request to the corresponding author.

Code Availability

Code is publicly available at `https://doi.org/10.5281/zenodo.18675630`.

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Author Contributions Statement

S.P. and V.P. formulated experiments. S.P. performed experiments. S.P. and V.P. analyzed the results. J.J. supervised and directed the project. All the authors contributed to writing the manuscript.

Competing Interests Statement

The authors declare no competing interests.

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